Geometry of Observation
Frozen Typed Corpus and Reproducibility Master
Eighteen reference modules, one scoped contract, and an immutable release map

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P12 corpus freeze v1.3.0 — 28 July 2026

Release status. This master freezes 18 normative modules with 158 component pages, 347 typed expressions, and zero GO Core findings. It is a provenance and reproducibility object. Concatenation does not turn the component results into one theorem, does not establish physical equivalence between application domains, and does not claim a new theory of general relativity, information theory, mechanics, quantum chemistry, accelerator physics, or astrodynamics.

Abstract

This release freezes the strict Geometry of Observation corpus after the P0–P11 migration sequence. The common mathematical discipline is not a universal physical dynamics but a typed separation of hidden state, invertible representation change, reduction, observation channel, sampling protocol, estimator, and declared invariant or recoverable target. The frozen object contains eighteen independently scoped reference modules, the GO Core v0.2 type and dimension contract, a machine-readable corpus ledger, an acyclic dependency ledger, canonical release metadata, exact component hashes, and executable validation. Every component PDF is embedded without textual revision; its original hash remains the authority for component identity. Short identifiers are document-scoped and receive fully qualified release names. Strong, model, hypothesis, diagnostic, and empirical claims remain separated. The master therefore records a clean corpus state while preserving the scientific boundaries of each module.

Аннотация. Релиз фиксирует строгий корпус геометрии наблюдения после миграций P0–P11. Общей структурой является не единая физическая динамика, а типизированное разделение скрытого состояния, обратимой смены представления, редукции, канала наблюдения, дискретизации, оценивателя и явно объявленного инварианта или восстанавливаемой величины. В master включены восемнадцать модулей без переписывания их содержимого, машинные реестры, зависимости, контрольные суммы и воспроизводимые проверки. Совпадение структуры между областями не трактуется как доказательство физической эквивалентности.

Contents

1 The frozen release object

Definition 1.1 (Corpus freeze)**.** Let

$$\mathfrak{F} = \big(C, L, O, D, S, H, M\big) \tag{1}$$

denote, respectively, the core contract, corpus ledger, canonical module order, dependency ledger, component source map, component hash map, and release metadata. A release is frozen when all seven entries are versioned and every file named by S has a verified digest in H. Two freezes are identical precisely when their canonical serializations and every referenced byte stream agree.

The baseline ledger is `go-corpus-ledgers@1.2.0`, with digest `3f50f7941f8549c4b9de626 fdb107431c07238582cdf8ebce90203500189f68e`. P12 does not mutate that ledger. Instead it adds a release-level ledger whose order and dependency semantics are explicit. A change to a component byte, ordering rule, dependency edge, author identity, licence, or release metadata requires a new release version. A filename change alone is not sufficient to create a new scientific object, while a changed byte stream cannot retain the old freeze identity.

Proposition 1.2 (Hash-addressed component identity)**.** *Assume collision resistance of SHA-256 for the finite release set. If two release records have the same ordered component paths, byte counts, and SHA-256 digests, then the probability of an undetected component substitution is negligible under the stated cryptographic assumption.*

Boundary 1.3. This is an integrity statement, not a mathematical proof that the scientific claims are true. Hash agreement proves preservation of the audited object; it does not replace peer review.

2 Common typed architecture

The minimal release-wide observation pattern is

$$\mathcal{X} \xrightarrow{\Phi_t} \mathcal{X}_t \xrightarrow{F_g} \mathcal{X}_t' \xrightarrow{R_\lambda} \mathcal{X}_\lambda \xrightarrow{\mathcal{Q}_\lambda} \mathcal{Y}_\lambda \xrightarrow{\mathcal{S}_{\Delta t,T}} \mathcal{Y}_\lambda^M \xrightarrow{\mathcal{A}} \mathcal{I}. \tag{2}$$

Here F_g is an invertible frame, gauge, coordinate, or basis change. The reduction R_λ, observation $\mathcal{Q}_\lambda$, and sampler $\mathcal{S}_{\Delta t,T}$ may lose information. The estimator $\mathcal{A}$ is a decision rule, not a physical transformation of the hidden system. A module may omit stages only by declaring the corresponding identity or noiseless map.

For a declared group G, protocol λ, and representation ρ, the release distinguishes

$$I(g\cdot x;\lambda) = I(x;\lambda) \qquad \text{(invariance)}, \tag{3}$$

$$\mathcal{Q}_\lambda(g\cdot x) = \rho(g)\mathcal{Q}_\lambda(x) \qquad \text{(covariance)}, \tag{4}$$

$$T(x) = \overline{T}(\mathcal{Q}_\lambda(x)) \qquad \text{(deterministic recoverability)}. \tag{5}$$

In a stochastic finite-alphabet setting, exact recoverability of a target $T(X)$ from Y is equivalently stated as

$$H(T(X) \mid Y) = 0 \tag{6}$$

under the declared probability law. None of these statements is well-typed without its transformation group, domain, protocol, and unit context.

Boundary 2.1 (No cross-domain identity claim). The recurrence of equation (2) in geometry, mechanics, spectroscopy, molecular inference, or satellite networks records a shared inference architecture. It does not identify their state spaces, Hamiltonians, field equations, probability laws, or empirical content.

3 Corpus layers and canonical order

The master order is semantic rather than chronological. It preserves every component identity while placing interfaces before the laboratories that use them.

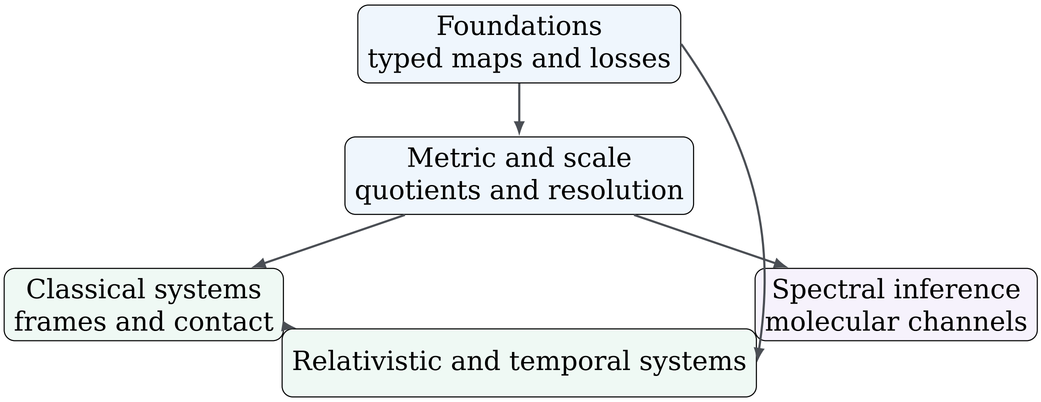

Table 1: Canonical semantic order of the frozen corpus. Counts are module-local.

No.	Layer	Module	v	pp.	Expr.	Claims
1	Foundation	Functional Interface GO-QS	0.2.0	12	3	2
2	Foundation	Tensorial Observation Geometry in GR	0.3.0	11	2	3
3	Information	Information-Theoretic Observation Geometry	0.2.0	9	9	9
4	Metric	Metric Entropy and Observational Entropy Defect	0.2.0	10	13	12
5	Metric	Distance and Scale Interface under Observation Maps	0.2.0	9	12	8
6	Scale	Planck-to-Cosmos Observation Rulers	1.1.0	12	19	10
7	Scale	Mandelbrot Rulers as Observation-Scale Geometry	1.1.0	11	12	9
8	Discrete geometry	Regular Polyhedra under Typed Observation Filters	1.1.0	7	42	10
9	Mechanics	Frames, Forces, Constraints, and Dissipation under Observation Maps	0.1.0	7	10	4
10	Mechanics	Foucault Networks on Celestial Bodies	1.1.0	6	7	4
11	Dynamics	Billiards as a Typed Geometry of Observation Laboratory	1.1.0	9	24	13
12	Contact mechanics	Bobsleigh Contact Geometry as a Typed Observation System	1.1.0	6	11	4
13	Mechanics	Roller-Coaster Geometry as a Typed Observation Laboratory	1.1.0	6	11	4
14	Contact mechanics	Gear Contact Geometry as a Typed Finite Observation System	1.1.0	9	20	9
15	Spectral	Conical Intersections as Typed Spectral Singularities under Observation Maps	1.1.0	8	36	11

No.	Layer	Module	v	pp.	Expr.	Claims
16	Molecular inference	Quantum Chemistry as a Typed Inference Stack under Observation Maps	1.1.0	9	47	11
17	Relativistic	Relativistic Beam Paths as Observation Geometry	1.3.0	9	16	4
18	Temporal networks	Satellite Networks under Typed Frames and Temporal Observation Channels	1.2.0	8	53	9

4 Inter-module dependency audit

Dependencies are typed. A *core-contract* edge is normative: the target ledger must validate against GO Core v0.2. An *extension-contract* edge binds a module to an auxiliary contract such as the SI–HEP passport or the mechanics interface. A *documented-interface* edge records an explicit source-level reference between modules. A *contextual* relation is excluded from the normative graph and cannot be used as a proof dependency.

Table 2: Explicit non-global dependency edges. The universal GO Core edge to all modules is omitted from the rows.

From	To	Edge kind
go-distance-scale-contract@0.3.0	Distance and Scale Interface under Observation Maps	extension contract
go-distance-scale-contract@0.3.0	Mandelbrot Rulers as Observation-Scale Geometry	extension contract
go-planck-cosmos-scale-contract@0.4.0	Planck-to-Cosmos Observation Rulers	extension contract
go-si-hep-quantity-passport@0.5.0	Relativistic Beam Paths as Observation Geometry	extension contract
go-frame-force-dissipation-contract@0.6.0	Frames, Forces, Constraints, and Dissipation unde...	extension contract
go-frame-force-dissipation-contract@0.6.0	Foucault Networks on Celestial Bodies	extension contract
go-frame-force-dissipation-contract@0.6.0	Bobsleigh Contact Geometry as a Typed Observation...	extension contract
go-frame-force-dissipation-contract@0.6.0	Roller-Coaster Geometry as a Typed Observation La...	extension contract
go-frame-force-dissipation-contract@0.6.0	Gear Contact Geometry as a Typed Finite Observati...	extension contract
go-gear-contact-contract@0.7.0	Gear Contact Geometry as a Typed Finite Observati...	extension contract

From	To	Edge kind
go-billiards-observation-contract@0.8.0	Billiards as a Typed Geometry of Observation Labo...	extension contract
go-conical-intersections-observation-contract@0.9.0	Conical Intersections as Typed Spectral Singulari...	extension contract
go-quantum-chemistry-observation-contract@1.0.0	Quantum Chemistry as a Typed Inference Stack unde...	extension contract
go-regular-polyhedra-observation-contract@1.1.0	Regular Polyhedra under Typed Observation Filters	extension contract
go-satellite-networks-observation-contract@1.2.0	Satellite Networks under Typed Frames and Tempora...	extension contract
Information-Theoretic Observation Geometry	Metric Entropy and Observational Entropy Defect	documented interface
Metric Entropy and Observational Entropy Defect	Distance and Scale Interface under Observation Maps	documented interface
Distance and Scale Interface under Observation Maps	Planck-to-Cosmos Observation Rulers	documented interface

The normative and documented-interface graph has 29 nodes and 36 edges. Its topological sort is unique only up to independent branches; acyclicity is the relevant release condition. Each edge endpoint resolves to a frozen module or a versioned contract, and each documented evidence fragment is checked against the cited source.

5 Namespaces and symbol discipline

Document-local identifiers are not silently promoted to global names. For document d and local identifier a, the master name is

$$\mathrm{qid}(d, a) = \mathtt{d::a}. \tag{7}$$

This operation is injective on ordered pairs. The audit found 7 repeated short identifiers across expression and claim registers; all become distinct after qualification. These overlaps are reuse of local vocabulary, not ledger collisions. Map identifiers and quantity identifiers are already globally unique in the frozen ledger.

Symbols remain scoped records, not bare glyphs. Thus G may denote a symmetry group in one module and the gravitational constant in another only when the scope and quantity identifier are carried with the expression. The master does not create a universal symbol table by typographic coincidence.

6 Claim stratification and scientific boundary

The corpus claim register contains 136 entries: 103 theorem/proposition/corollary records, 14 models, 11 hypotheses, 7 diagnostics, and 1 empirical record. A strong label is accepted by the release only when the local ledger supplies the hypotheses and the source states the result in its declared scope. P12 does not strengthen a component claim.

Layer	Release interpretation
Definition	Fixes notation or an object; no empirical claim.
Theorem, proposition, corollary	Mathematical statement under the local hypotheses.
Model	Declared idealization or computational benchmark.
Hypothesis	Unproved assumption or proposed condition.
Diagnostic	Protocol-dependent summary; not an invariant unless separately proved.
Empirical	Externally measured input with provenance and uncertainty requirements.

Boundary 6.1 (Master claim firewall). The release does not claim a theory of everything, a derivation of Standard Model parameters, a new electronic-structure method, a new ergodicity theorem, a new classification of regular polyhedra, or an operational orbit-determination system. Its defensible contribution is the typed audit infrastructure and the module-specific corrections recorded in the appended documents.

7 Metadata normalization and provenance

The canonical release author is **Stassis Stashkevichyus**, identified by ORCID 0009-0000-2294-705X. Earlier component PDFs contain the historical display strings *Stassis Research Program* or *Stas, Independent Research Program*. Their bytes are preserved; the container metadata supplies one normalized creator record without pretending that the old PDF metadata was rewritten.

No DOI is fabricated in this freeze. The release metadata is prepared for one corpus-level archival record; component modules are not assigned separate placeholder identifiers. If a DOI is minted, the immutable release metadata must be updated in a successor version or by an archive-controlled identifier field that does not alter the deposited files.

The root citation file follows Citation File Format 1.2.0. The Zenodo metadata file follows the documented GitHub-release override structure. The release is marked CC BY-NC-ND 4.0 as directed by the author. This licence permits non-commercial redistribution of unadapted material with attribution; it is not an open-source software licence and does not permit distribution of modified versions.

8 Validation and reproducibility

The P12 gate requires all of the following:

1. exact resolution of all 18 PDF and source paths;
2. agreement of component page counts, byte counts, and SHA-256 digests with the freeze ledger;
3. a clean GO Core replay: 18 PASS/0 FAIL/0 BLOCKED;
4. uniqueness of document IDs and fully qualified local IDs;
5. an acyclic, endpoint-complete dependency graph with verified source evidence;
6. successful execution of the P1–P11 release validators;

7. A4 geometry, no encryption or form fields, embedded fonts, and extractable text for every component and the master;
8. page-content preservation between each source PDF and its corresponding master page interval;
9. valid citation, archival, licence, and checksum records;
10. a deterministic ZIP archive with fixed timestamps, sorted members, and byte-identical rebuild.

The master contains 8 release-front-matter pages followed by the 158 unchanged component pages, for a total of 166 pages. PDF bookmarks and the external page map identify every component boundary. The component PDFs, not the master page numbering, remain the primary unit of scientific citation.

9 How to cite and review this object

Scientific review should proceed module by module. A reviewer should first inspect the claim register and protocol fields, then the local proofs or counterexamples, and finally the numerical and release checks. A criticism of one application module does not automatically invalidate the typed core; conversely, a clean core lint does not validate an application theorem whose hypotheses are false.

Until an archival DOI exists, cite the title, author, ORCID, version, date, and the SHA-256 digest recorded in `SHA256SUMS_v1_3.txt`:

> Stassis Stashkevichyus, *Geometry of Observation: Frozen Typed Corpus and Reproducibility Master*, P12 corpus freeze v1.3.0, 28 July 2026, ORCID 0009-0000-2294-705X.

10 Release conclusion

P12 closes the migration sequence at the corpus level. The defensible result is an immutable, typed, hash-addressed collection of eighteen reference modules with explicit claim boundaries and reproducible validation. The release is suitable for archival provenance and external module-by-module review. It is not, by itself, evidence that the umbrella programme has become a single new physical theory.